

Eval整理

Agent

Code Agent

- McEval: Massively Multilingual Code Evaluation
<https://arxiv.org/abs/2406.07436>
- M2rc-Eval: Massively Multilingual Repository-level Code Completion Evaluation
<https://arxiv.org/abs/2410.21157>
- FullStack Bench: Evaluating LLMs as Full Stack Coders
<https://arxiv.org/abs/2412.00535>
- SWE-Compass: Towards Unified Evaluation of Agentic Coding Abilities for Large Language Models
<https://arxiv.org/abs/2511.05459>
- ArtifactsBench: Bridging the Visual-Interactive Gap in LLM Code Generation Evaluation
<https://arxiv.org/abs/2507.04952>
- NL2Repo-Bench: Towards Long-Horizon Repository Generation Evaluation of Coding Agents
<https://arxiv.org/pdf/2512.12730>
- Multi-Docker-Eval: A Shovel of the Gold Rush' Benchmark on Automatic Environment Building for Software Engineering
<https://arxiv.org/pdf/2512.06915>
- CodeTracer: Towards Traceable Agent States (CodeTraceBench)
<https://arxiv.org/pdf/2604.11641>
- WebCompass: Towards Multimodal Web Coding Evaluation for Code Language Models



WebCompass: Towards Multimodal Web Coding Evaluation for Code Language Models

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Abstract

Large language models are rapidly evolving into interactive coding agents capable of end-to-end web coding, yet existing benchmarks evaluate only program slices of

or end-to-end web coding, yet existing benchmarks evaluate only narrow slices of this capability—typically text-conditioned generation with static-correctness metrics—leaving visual fidelity, interaction quality, and codebase-level reasoning largely unmeasured. We introduce **WebCompass**, a comprehensive, multimodal benchmark that provides a *unified lifecycle evaluation* of web engineering capability. Recognizing that real-world web coding is an iterative cycle of generation, editing, and repair, WebCompass spans three input modalities (**text, image, and video**) and three tightly coupled task types (**generation, editing, and repair**), yielding seven complementary task categories that closely mirror professional workflows. Through a multi-stage, human-in-the-loop pipeline, we curate high-quality instances covering 15 generation domains, 16 editing operation types, and 11 repair defect types, each annotated at Easy/Medium/Hard difficulty levels. On the evaluation side, we adopt a *checklist-guided LLM-as-a-Judge* protocol for editing and repair, and propose a novel **Agent-as-a-Judge** paradigm for generation that autonomously executes generated websites in a real browser, explores interactive behaviors via the Model Context Protocol (MCP), and iteratively synthesizes targeted test cases—closely approximating human acceptance testing. We evaluate a diverse set of representative closed-source and open-source models and observe that: (1) closed-source models remain substantially stronger and more balanced; (2) editing and repair exhibit distinct difficulty profiles, with repair preserving interactivity better but remaining execution-challenging; (3) aesthetics is the most persistent bottleneck, especially for open-source models; and (4) framework choice materially affects outcomes, with Vue consistently challenging while React and Vanilla/HTML perform more strongly depending on task type. All benchmark data¹, evaluation code², and supplementary artifacts³ are publicly available.

1 Introduction

Large Language Models (LLMs) have rapidly evolved from passive code assistants into interactive coding agents capable of implementing substantial software changes from natural-language instructions (Yang et al., 2024b; Wang et al., 2024; Cognition AI, 2024). This progress is especially evident in **web development**, where outputs can be directly executed, visually inspected, and iteratively refined. A growing body of work has proposed benchmarks that span different task types and input modalities for web coding (Table 1).

Yet evaluating web coding is fundamentally different from evaluating traditional code generation. Success depends not only on functional correctness, but also on visual fidelity, interaction behavior, responsiveness, accessibility, and overall user experience. These aspects are difficult to capture with standard code-centric metrics such as pass@k on HumanEval (Chen et al., 2021) or unit-test pass rates on SWE-Bench (Jimenez et al., 2023), which focus on algorithmic correctness or repository-level bug fixing rather than interactive front-end applications.

To address this gap, we introduce **WebCompass**, a unified multimodal benchmark and evaluation framework for web coding. WebCompass spans **text, image, and video** inputs, covers **generation, editing, and repair** tasks, and adopts **task-aware evaluation** tailored to each setting. For editing and repair, we use a *checklist-guided LLM-as-a-Judge* protocol (Zheng et al., 2023), which is well suited to

¹<https://huggingface.co/datasets/NJU-LINK/WebCompass>

²<https://github.com/NJU-LINK/WebCompass>

³<https://nju-link.github.io/WebCompass/>

Search Agent

- How Far Are We from Genuinely Useful Deep Research Agents?
<https://arxiv.org/abs/2512.01948>
- DR3-Eval: Towards Realistic and Reproducible Deep Research Evaluation
<https://arxiv.org/abs/2604.14683>
- ACADREASON: Exploring the Limits of Reasoning Models with Academic Research Problems
<https://arxiv.org/pdf/2510.11652?>
- TVIR <https://github.com/NJU-LINK/TVIR>

TVIR: Building Deep Research Agents Towards Text–Visual Interleaved Report Generation

Anonymous ACL submission

Abstract

Deep Research Agents have shown strong capability in multi-step information retrieval, reasoning, and long-form report generation, but existing benchmarks and systems remain predominantly text-centric, with limited evaluation of whether visual elements are factually reliable and well aligned with the surrounding analysis. To address this gap, we introduce **TVIR** (Text–Visual Interleaved Report Generation), which includes **TVIR-BENCH**, a benchmark of 100 expert-curated multimodal deep research tasks that require visual elements to serve specific analytical sub-goals, and **TVIR-AGENT**, a hierarchical multi-agent framework for constructing structured outlines, retrieving images, generating charts with traceable sources, and composing reports through context-aware sequential writing. We further develop a dual-path evaluation framework that combines **Textual Assessment** and **Visual Assessment**. Experiments across nine deep research systems show that TVIR-AGENT achieves strong overall performance, underscoring the importance of explicit multimodal design and evaluation for evidence-driven report generation.

1 Introduction

Recent advances in large language models (LLMs) (OpenAI et al., 2024; DeepSeek-AI et al., 2025) have catalyzed the emergence of Deep Research Agents (DRAs) (Zheng et al., 2025; Coelho et al., 2025; Cai et al., 2026; Tongyi DeepResearch Team et al., 2025), which aim to autonomously conduct multi-step information retrieval, reasoning, and generation to produce comprehensive research reports. These systems have shown promising capability in long-context planning, citation grounding, and analytical writing, positioning them as potential assistants for complex professional decision-making in domains such as policy analysis, finance, and scientific research. As a result, a growing body of benchmarks and systems (Shi et al., 2025) has

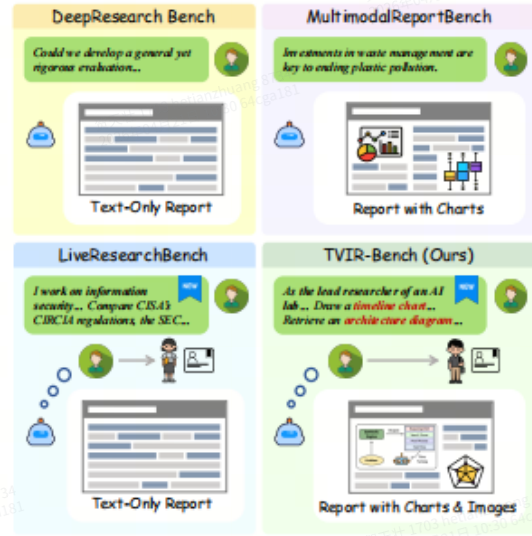


Figure 1: Comparison of representative deep research benchmarks. Existing benchmarks mainly focus on text-only or weakly multimodal reports, whereas TVIR-BENCH requires text–visual interleaved reports with semantically grounded charts and retrieved images.

focused on evaluating end-to-end deep research workflows rather than isolated questions.

Despite this progress, as shown in Figure 1, existing deep research paradigms remain predominantly text-centric. Most benchmarks and agent frameworks evaluate success based on textual coherence, depth, and citation support, while overlooking a critical characteristic of real-world professional reports: the integration of visual evidence. In practice, high-quality research outputs rarely rely on text alone. Instead, they interleave narrative analysis with charts, diagrams, and images that serve as evidential artifacts—supporting claims, revealing trends, and enabling rapid sense-making. When visual elements are present in current systems, they are often treated as decorative supplements rather than first-class reasoning components, with limited

Multimodal Agent

- WorldTravel: A Realistic Multimodal Travel-Planning Benchmark with Tightly Coupled Constraints <https://arxiv.org/pdf/2602.08367>
- MM-BrowseComp: A Comprehensive Benchmark for Multimodal Browsing Agents <https://arxiv.org/pdf/2508.13186>

MLLM Benchmark

Image Understanding

- II-Bench: An Image Implication Understanding Benchmark for Multimodal Large Language Models <https://arxiv.org/abs/2406.05862>
- LIME: Less Is More for Evaluation of MLLMs <https://arxiv.org/abs/2409.06851>
- Can MLLMs Understand the Deep Implication Behind Chinese Images? <https://arxiv.org/abs/2410.13854>

Long Video Understanding

- MT-Video-Bench: A Holistic Video Understanding Benchmark for Evaluating Multimodal LLMs in Multi-Turn Dialogues <https://arxiv.org/abs/2510.17722>
- ScaleLong: A Multi-Timescale Benchmark for Long Video Understanding <https://arxiv.org/pdf/2505.23922>
- MVU-Eval: Towards Multi-Video Understanding Evaluation for Multimodal LLMs <https://arxiv.org/abs/2511.07250>
- IV-Bench: A Benchmark for Image-Grounded Video Perception and Reasoning in Multimodal LLMs <https://arxiv.org/pdf/2504.15415>

Video Captioning

- ViDiC: Video Difference Captioning <https://arxiv.org/abs/2512.03405>
- IF-VidCap: Can Video Caption Models Follow Instructions? <https://arxiv.org/pdf/2510.18726>
- VidCapBench: A Comprehensive Benchmark of Video Captioning for Controllable Text-to-Video Generation <https://arxiv.org/abs/2502.12782>

Omni Understanding

- OmniBench: Towards The Future of Universal Omni-Language Models
<https://arxiv.org/abs/2409.15272>
- OmniVideoBench: Towards Audio-Visual Understanding Evaluation for Omni MLLMs
<https://arxiv.org/abs/2510.10689>
- OmniCap-IF: Benchmarking and Improving Instruction Following Abilities for Omni-Video Captioning



3199_OmniCap_IF_Benchmarking
_a.pdf
18.65MB



AIGC

T2AV-Compass: Towards Unified Evaluation for Text-to-Audio-Video Generation
<https://arxiv.org/pdf/2512.21094>

LLM Benchmark

Instruction Following

- MT-Bench-101: A Fine-Grained Benchmark for Evaluating Large Language Models in Multi-Turn Dialogues
<https://arxiv.org/abs/2402.14762>
- Inverse IFEval: Can LLMs Unlearn Stubborn Training Conventions to Follow Real Instructions?
<https://arxiv.org/abs/2509.04292>

Long Context

- Can Large Language Models Detect Errors in Long Chain-of-Thought Reasoning?
<https://arxiv.org/abs/2502.19361>
- HelloBench: Evaluating Long Text Generation Capabilities of Large Language Models
<https://arxiv.org/abs/2409.16191>
- Long-form RewardBench: Evaluating Reward Models for Long-form Generation
<https://arxiv.org/abs/2603.12963>

Math

- ConceptMath: A Bilingual Concept-Wise Benchmark for Measuring Mathematical Reasoning of Large Language Models
<https://arxiv.org/abs/2402.14660>
- CriticLean: Critic-Guided Reinforcement Learning for Mathematical Formalization (CriticLeanBench) <https://arxiv.org/abs/2507.06181>

Reasoning

- KOR-Bench: Benchmarking Language Models on Knowledge-Orthogonal Reasoning Tasks
<https://arxiv.org/abs/2410.06526>
- KORgym: A Dynamic Game Platform for LLM Reasoning Evaluation
<https://arxiv.org/abs/2505.14552>
- TableBench: A Comprehensive and Complex Benchmark for Table Question Answering
<https://arxiv.org/abs/2408.09174>
- MTU-Bench: A Multi-granularity Tool-Use Benchmark for Large Language Models
<https://arxiv.org/abs/2410.11710>

Knowledge

- SuperGPQA: Scaling LLM Evaluation across 285 Graduate Disciplines
<https://arxiv.org/abs/2502.14739>
- Chinese SimpleQA: A Chinese Factuality Evaluation for Large Language Models
<https://arxiv.org/abs/2411.07140>
- OWL: A Large Language Model for IT Operations
<https://arxiv.org/abs/2309.09298>

Safety

- Chinese SafetyQA: A Safety Short-form Factuality Benchmark for Large Language Models
<https://arxiv.org/abs/2412.15265>
- SafeDialBench: A Fine-Grained Safety Evaluation Benchmark for Large Language Models in Multi-Turn Dialogues with Diverse Jailbreak Attacks
<https://arxiv.org/abs/2502.11090>
- USB: A Comprehensive and Unified Safety Evaluation Benchmark for Multimodal Large Language Models <https://arxiv.org/abs/2505.23793>

Role-playing

- RoleLLM: Benchmarking, Eliciting, and Enhancing Role-Playing Abilities of Large Language Models

<https://arxiv.org/abs/2310.00746>

- RoleAgent: Building, Interacting, and Benchmarking High-quality Role-Playing Agents from Scripts

<https://openreview.net/forum?id=hORTHzt2cE>